# 1. Dataset Information

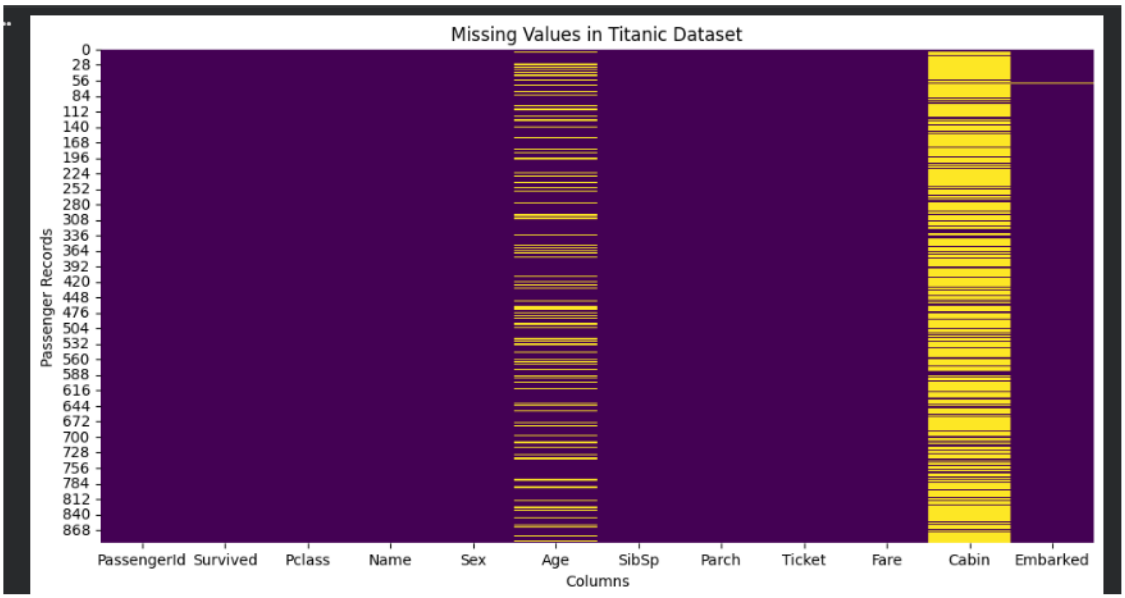
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype  
---  ------       --------------  -----  
 0   PassengerId  891 non-null    int64  
 1   Survived     891 non-null    int64  
 2   Pclass       891 non-null    int64  
 3   Name         891 non-null    object 
 4   Sex          891 non-null    object 
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64  
 7   Parch        891 non-null    int64  
 8   Ticket       891 non-null    object 
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object 
 11  Embarked     889 non-null    object 
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

# 2. Missing Values Before Cleaning

```
print("Missing Values in Each Column:")

print(df.isnull().sum())
```

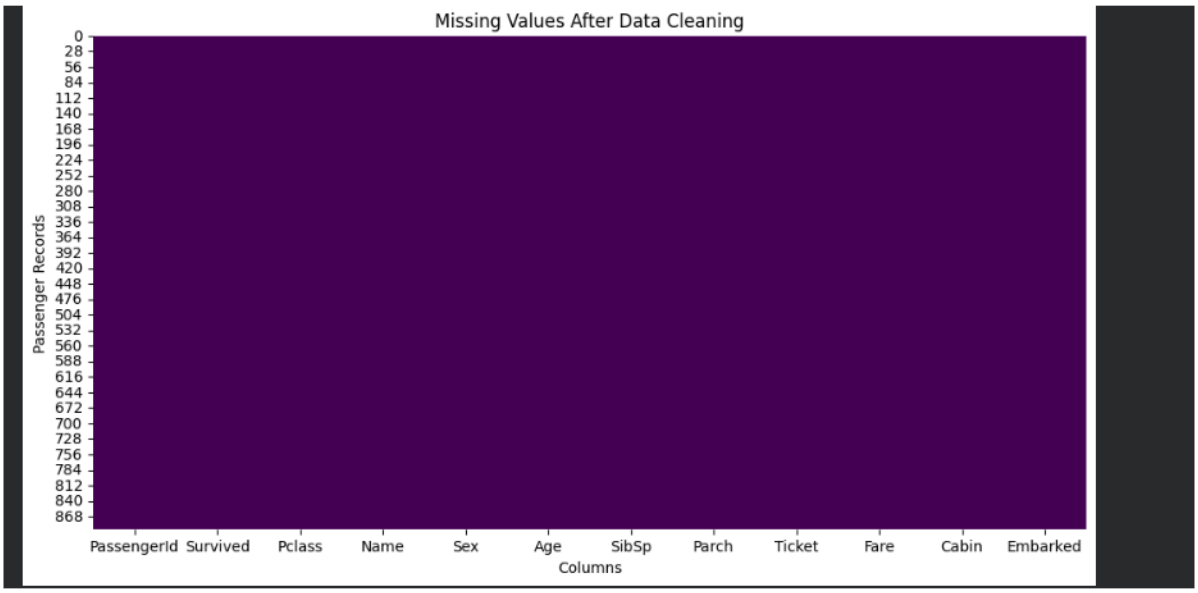
```
Missing Values in Each Column:
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```



# 3. Missing Values After Cleaning

```
Missing Values After Cleaning:
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
Age            0
SibSp          0
Parch          0
Ticket         0
Fare           0
Cabin          0
Embarked       0
dtype: int64

Total Missing Values: 0
```



# 4. Categorical Encoding

```
print(df[["Sex", "Sex_Encoded"]].head())
```

```
      Sex  Sex_Encoded
0    male            1
1  female            0
2  female            0
3  female            0
4    male            1
```

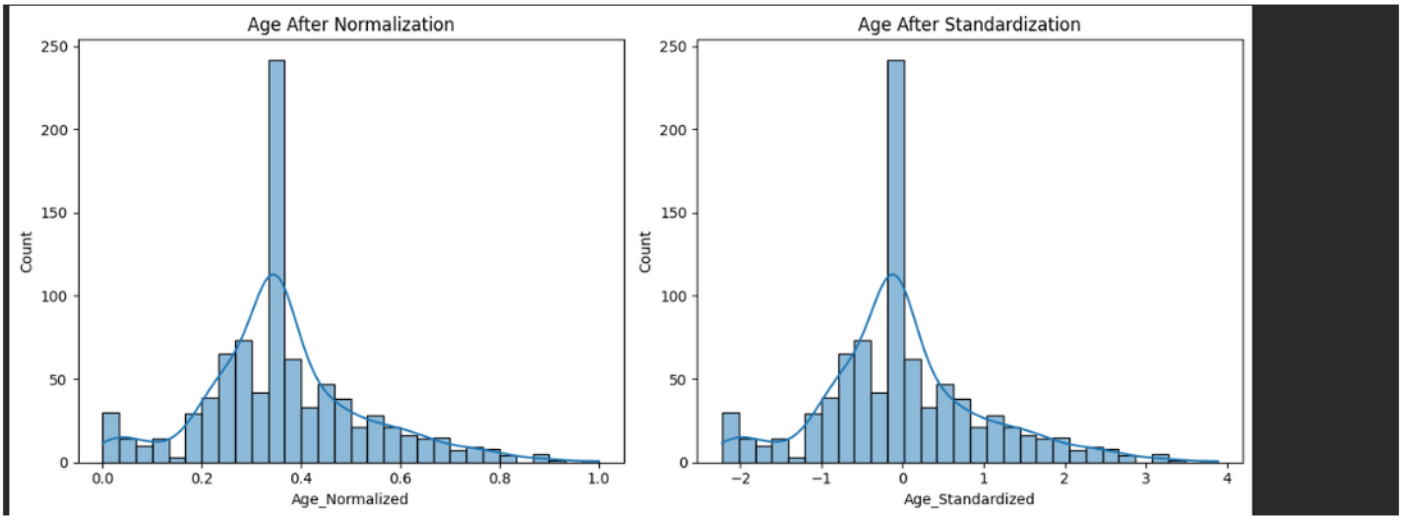
```
Encoded Dataset:
      Sex  Sex_Encoded
0    male            1
1  female            0
2  female            0
3  female            0
4    male            1

One-Hot Encoded Columns:
   Embarked_C  Embarked_Q  Embarked_S
0           0           0           1
1           1           0           0
2           0           0           1
3           0           0           1
4           0           0           1
```

# 5. Normalization & Standardization

```
Age Feature Comparison:
    Age  Age_Normalized  Age_Standardized
0  22.0        0.271174         -0.565736
1  38.0        0.472229          0.663861
2  26.0        0.321438         -0.258337
3  35.0        0.434531          0.433312
4  35.0        0.434531          0.433312

Fare Feature Comparison:
      Fare  Fare_Normalized  Fare_Standardized
0   7.2500         0.014151          -0.502445
1  71.2833         0.139136           0.786845
2   7.9250         0.015469          -0.488854
3  53.1000         0.103644           0.420730
4   8.0500         0.015713          -0.486337
```

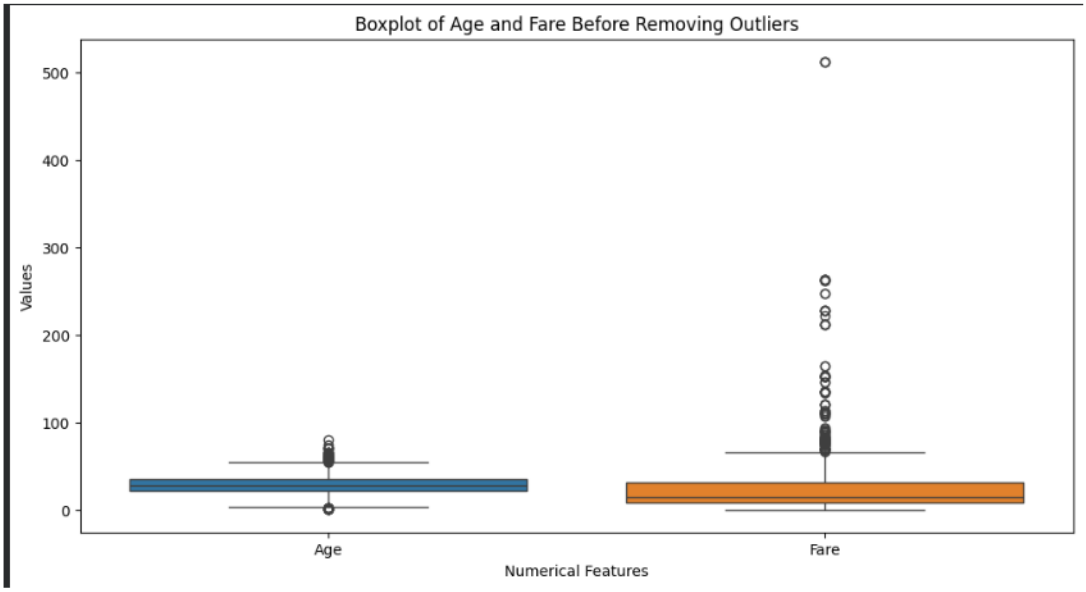



```
Normalized Age Range:
0.0 1.0

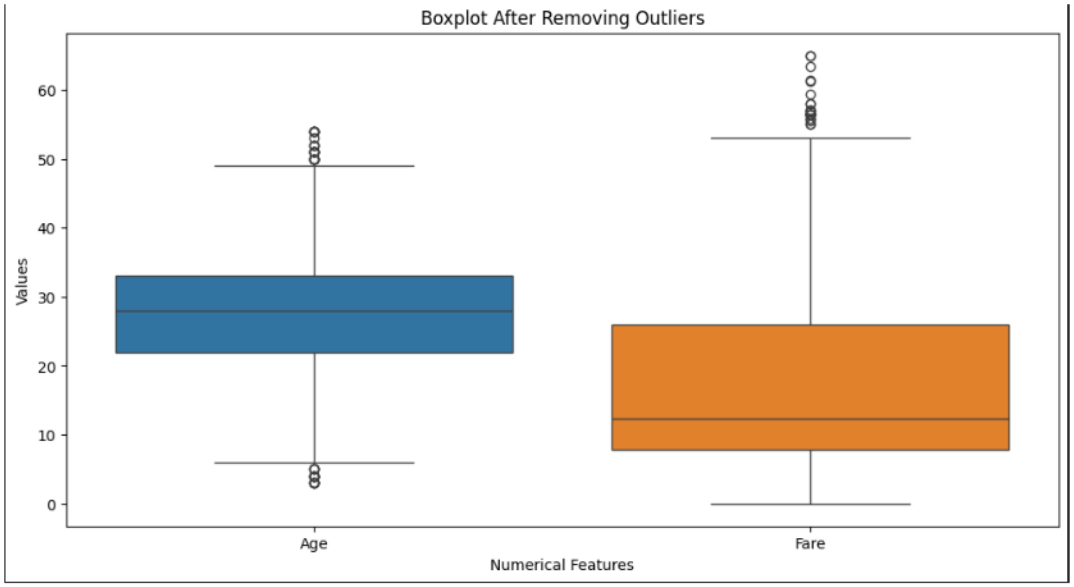
Standardized Age Mean:
2.272779794518839e-16

Standardized Age Standard Deviation:
1.000561640033049
```

# 6. Outliers Before Removal



# 7. Outliers After Removal




```
Before Outlier Removal:
Rows: 891

After Outlier Removal:
Rows: 721

Total Rows Removed:
170
```

# 8. Final Cleaned Dataset

```
from google.colab import files

files.download("Titanic_Cleaned_Dataset.csv")
```